

findings

§11.4.55 Sweep Remediation Recipes — 2026-06-22

Scope: READ-ONLY remediation report for the 6 failing gates of `scripts/verify-all-constitution-rules.sh` (G2 already fixed this session). No source/governance files were edited or committed. (One transient G12 in-place regen was performed to MEASURE the diff, then immediately reverted — working tree is back to its pre-investigation state apart from the pre-existing parallel-session `semgrep` changes.)

Live sweep verdict (this run): Gates run: 14 · Failures: 6 **G2 now PASS** (“zero production bluff markers”) — disproves the stale “7-fail” figure; the live count is **6/14 failing**.

Gate	Anchor	One-line verdict	Pre-existing?	Safe to auto-fix?
G1	§11.9 + CONST-047 cascade	213 cascade failures (63 PASS); the queued 62/64-repo fleet 142 → 166 backfill	pre-existing (queued)	NO — operator-scoped fleet cascade (do not run inline)
G7	§11.4.83 docs/qa	117 feature commits lack <code>docs/qa/<run-id>/</code>	pre-existing debt (116) + 1 this-session HEAD	PARTLY — [no-qa-evidence] is not retro-applicable; needs operator decision
G8	§11.4.90 Obsolete-Details	1 violation: HXC-044 Reason out-of-vocab + Triple-check token malformed	pre-existing (2026-06-09)	NO — closed-vocab Reason needs operator decision
G12	§11.4.12 summary freshness	2 summaries stale by ONLY the Last regenerated date stamp	pre-existing	YES — pure regen, no content change
G13		22/72 operator docs lack a <code>##</code>		NO — §11.4.99 requires live

Gate	Anchor	One-line verdict	Pre-existing?	Safe to auto-fix?
	§11.4.99 Sources-verified	Sources verified footer	pre-existing (docs dated 06-14..06-22)	WebFetch cross-ref, not a blind append
G14	§11.4.106 docs_chain	3 contexts STALE (issues/fixed/ governance + new semgrep_status)	pre-existing + parallel- session	YES (mostly downstream of G12) — docs_chain sync --all

G1 — Governance cascade (§11.9 + CONST-047) — CONFIRMED queued fleet cascade

Violation count: 213 FAIL / 63 PASS (/tmp/g1-cascade.out, 317 KB). **Distinct owned submodules failing:** 64, plus 6 root carriers (root/CLAUDE.md lags by 1 anchor = §11.4.166; AGENTS/QWEN/GEMINI lag from §11.4.159; CRUSH/CONSTITUTION lag from §11.4.142).

Root cause: the constitution submodule was bumped to 4ff4985 (adds §11.4.157 GEMINI-lockstep, §11.4.166 semgrep). Per CONST-047 the new anchor band §11.4.142 → 166 must propagate (verbatim or by ID) into every owned submodule's CLAUDE/AGENTS/CONSTITUTION trio + the root carriers. None of the 207 owned carrier files (64 submodules) carry the band yet.

Pre-existing? PRE-EXISTING / queued. This is the cross-cutting fleet cascade explicitly out-of-scope for this task. **NOT re-investigated** per instructions — confirmed it is the 62-repo (here: 64-submodule) fleet 142 → 166 backfill.

Fix recipe (operator-scoped — do NOT run inline):

```
# Cascade the new anchor band into every owned carrier + root mirrors:
bash scripts/propagate-governance.sh           # root carriers
(CLAUDE/AGENTS/QWEN/GEMINI/CRUSH/CONSTITUTION)
bash scripts/backfill_anchor_cascade.sh         # owned-submodule
trio backfill (142→166)
bash scripts/verify-governance-cascade.sh       # re-verify →
expect 0 failures
# then commit+push EACH owned submodule (no force, §11.4.113) +
bump pointers.
```

Effort: LARGE (64 submodules × 3 files, each committed+pushed to all upstreams, then pointer-bumped). **Safe to auto-fix?** NO — multi-repo, multi-upstream, operator-gated.

G7 — §11.4.83 docs/qa end-user evidence — 117 violations

Violation count: 117 (/tmp/g7-qa.out). Scope = ed84f90e..HEAD (baseline 2026-05-28; 453 commits in scope). **Sample (newest 3):** - ab5d2ad2 chore(governance): G2 smoke false-positive fix ... (← THIS session, HEAD) - 1771c855 fix(cli): add Xiaomi MiMo models to -list-models output - 759e48c8 feat(llm): add ASR and TTS endpoint support to Xiaomi provider Span: 2026-06-10 (bd3adb7f) → 2026-06-22 (ab5d2ad2).

Root cause: the gate heuristic flags any feat(/fix(commit touching shippable code that has no matching docs/qa/<run-id>/ directory. 116 of the 117 are PRE-EXISTING feature commits (Xiaomi provider, §11.4.118 discovery rounds, HXC-NNN fixes) that landed without a QA transcript dir; 1 (ab5d2ad2) is this session's governance commit that tripped the chore(+code heuristic. docs/qa/ exists with 61 run-id dirs — the tree is real, just not populated for these 117 commits.

Pre-existing? YES — 116/117 pre-existing debt across 453 commits.

Fix recipe (two legitimate paths per the gate's own message): 1. For genuine feature commits → author the missing evidence: mkdir -p docs/qa/<run-id>/ && <add bidirectional transcript + materials> (one dir per shipped feature; see docs/qa/README.md). This is the §11.4.83-compliant path but is heavy (117 transcripts). 2. For non-feature changes that tripped the heuristic (e.g. ab5d2ad2 governance) → the gate is git-history-based and the annotation is [no-qa-evidence] in the COMMIT MESSAGE. It is **not retro-applicable** to already-pushed commits without history rewrite (forbidden, §11.4.113). Realistic remediation: a one-time baseline bump — ed84f90e..HEAD → set the gate baseline to a recent commit so the pre-baseline debt is exempted (same mechanism the gate already uses), OR add a docs/qa/<run-id>/ for each future feature going forward.

Effort: LARGE if authoring 117 transcripts; SMALL if operator approves a baseline bump for the pre-existing cohort + commit-message discipline forward. **Safe to auto-fix?** NO — requires operator decision on baseline-bump vs 117-transcript backfill; cannot fabricate transcripts (would itself be a bluff).

G8 — §11.4.90 Obsolete-Details — 1 violation (HXC-044)

Violation (/tmp/g8-obs.out):

FAIL §11.4.90 [docs/Fixed.md] ## HXC-044 — internal/cognee: AMD-GPU rocm-s
JSON parser ... — missing/invalid sub-fact(s): Reason(closed-vocab) Trip

Current line (docs/Fixed.md:294):

****Obsolete-Details:**** Since: 2026-06-09; Reason: not-reproducible; Superse

Root cause (TWO defects, both mechanical): 1. Reason: not-reproducible is NOT in the §11.4.90 closed vocabulary. The gate (scripts/gates/obsolete_details_gate.sh:28,55) accepts ONLY: superseded-by-design-change | superseded-by-later-mandate | feature-removed | duplicate-of | unsupported-topology. 2. The gate regex requires the literal token Triple-check: followed by content (/Triple-check: []*[^]/, line 57). The doc writes Triple-check evidence: — a SPACE not a COLON after Triple-check, so it never matches. (Note: there is NO closed-vocab value meaning “not reproducible”; the item is conceptually a non-reproducible env artefact.)

Pre-existing? YES — added 2026-06-09 (6f1c2ca5), ~13 days old; the gate has flagged it every run since.

Fix recipe: edit docs/Fixed.md:294 (and mirror the parallel pipe-row at docs/Fixed.md:13). The Triple-check token fix is unambiguous; the Reason needs an operator/judgment mapping into the closed vocab. Closest fits: - duplicate-of — NO (not a duplicate). - unsupported-topology — DEFENSIBLE (the failure was a worktree/topology artefact, “10/10 main-tree PASS”; the closure note literally says “worktree env artifact”). Proposed compliant line (operator to confirm the Reason choice):

****Obsolete-Details:**** Since: 2026-06-09; Reason: unsupported-topology; Sup

ALTERNATIVE (cleaner, but needs constitution change): petition to ADD not-reproducible to the §11.4.90 closed vocab (the vocab is “open to additions, never contraction” per the anchor) + the gate regex — that is a constitution-submodule edit (CONST-049 workflow), heavier.

Effort: SMALL (one/two-line doc edit) once Reason is chosen. **Safe to auto-fix?** NO — the Reason value is a judgment call (operator must pick unsupported-topology vs a vocab extension). The Triple-check: token fix alone is safe but insufficient.

G12 — §11.4.12 summary freshness — 2 stale summaries

Violation (/tmp/g12-summary.out): both docs/Issues_Summary.md (last regen 2026-06-16) and docs/Fixed_Summary.md (last regen 2026-06-17) are STALE vs a

fresh regeneration. **Measured diff = exactly 1 line each — ONLY the *Last regenerated:** <date> ...* stamp (2026-06-16/17 → 2026-06-22). Zero content/row changes. (Verified by running the generators to a temp + diff; then reverted.)

Root cause: the generators stamp the regen date; the gate regenerates with TODAY's date and byte-compares, so any day on which a tracker doc was touched without re-running the generators leaves the stamp stale. After in-place regen, `generate_{issues,fixed}_summary.sh --check` both report **PASS**.

Pre-existing? YES — stamps are 5-6 days old; trivial drift.

Fix recipe (CONFIRMED resolves the gate):

```
bash scripts/generate_issues_summary.sh      # in-place write
      (default mode)
bash scripts/generate_fixed_summary.sh      # in-place write
      (default mode)
# verify: both --check report "PASS – summary in sync"
bash scripts/generate_issues_summary.sh --check
bash scripts/generate_fixed_summary.sh --check
# then also regen exports (G14 depends on this – see below):
bash scripts/regenerate-tracker-exports.sh  # or `docs_chain
      sync --all --root .`
```

Effort: TRIVIAL (2 commands). **Safe to auto-fix?** YES — pure mechanical projection, no content change beyond the date stamp.

G13 — §11.4.99 Sources-verified footers — 22 missing

Violation (/tmp/g13-sv.out): 50/72 operator-facing docs footered (69%); 22 lack a `## Sources verified <date>: <urls> footer`. Gate is `--enforce` (100% required per HXC-030 closure). Sample of the 22: `docs/CODEGRAPH.md`, `docs/OPENDESIGN.md`, all 11 `docs/helixtrack/*.md` (API_REFERENCE, ARCHITECTURE, DATABASE_SCHEMA, DEPLOYMENT_GUIDE, FEATURE_INVENTORY, GAP_ANALYSIS, IMPLEMENTATION_PLAN, SECURITY_GUIDE, TEST_STRATEGY, UI_UX_FINDINGS, USER_MANUAL), 5 `docs/features/inventory/*.md`, `docs/features/xiaomi-status_summary.md`, `docs/features/_status_header.md`, `docs/plans/POWER_FEATURES_PORTING_PLAN.md`, `docs/SESSION_STATUS_2026-06-21.md`.

Footer format (from compliant docs, e.g. docs/ARCHITECTURE.md:265):

```
## Sources verified 2026-06-04: <url1>, <url2>, <repo cross-refs>
```

Root cause: these 22 operator-facing docs landed (06-14..06-22, all TRACKED, all in PRIOR commits) without the §11.4.99 footer; the enforcing gate flags them. Most are NEW doc subtrees (helixtrack/, features/inventory/, CODEGRAPH/ OPENDESIGN added 06-22 11:26 — predate the sweep).

Pre-existing? YES — all 22 are committed before this session’s sweep run.

Fix recipe (per §11.4.99 — NOT a blind append): for EACH of the 22: 1. Identify the doc’s authoritative sources (the libraries/services/repo state it documents). 2. WebFetch the LATEST official docs + cross-reference every instruction. 3. Append:

Sources verified 2026-06-22: <fetched-source-urls>, <repo cross-refs>

1. Record negative findings explicitly. 5. Cite the footer in the commit. For internal/repo-derived docs (helixtrack ARCHITECTURE/DATABASE_SCHEMA, features/inventory) the “sources” are the repo files cross-referenced (the ARCHITECTURE.md footer precedent lists .gitmodules, go.mod, ls outputs).

Effort: MEDIUM-LARGE (22 docs × WebFetch-cross-ref). Parallelisable by subagent (one per doc cluster: helixtrack, features, top-level). **Safe to auto-fix?** NO — §11.4.99 explicitly forbids footering from memory; each requires a live source fetch + cross-reference. A blind echo '## Sources verified ...' append would itself be a §11.4.99 bluff.

G14 — §11.4.106 docs_chain verify — 3-4 STALE contexts

Violation (/tmp/g14-verify.log):

```
fixed          STALE: [fixed_summary]
governance     STALE: [claude_html crush_html]
issues         STALE: [issues_html issues_pdf issues_summary summary_html sum
```

(A direct docs_chain verify --all also shows semgrep_status STALE: [semgrep_status_html semgrep_status_pdf] — from the parallel-session untracked semgrep_status.yaml context.)

Root cause: content-hash drift across docs_chain contexts: - fixed/issues summary nodes = DOWNSTREAM of G12 (same stale summaries) + their .html/.pdf exports never regenerated after the last tracker edit. - governance (claude_html, crush_html) = CLAUDE.md/CRUSH.md changed (the §11.4.166 governance work) without re-exporting their HTML siblings. - semgrep_status = new parallel-session context not yet synced. Engine is available 3 ways:

docs_chain on PATH (~/.local/bin/docs_chain), sibling ../docs_chain/cmd/docs_chain, or .docs_chain/bin/ (absent).

Pre-existing? YES (summary + governance drift) + PARTLY parallel-session (semgrep_status context).

Fix recipe (the gate prints it verbatim):

```
docs_chain sync --all --root .      # atomically propagates every
                                   context, updates state.json
docs_chain verify --all --root .    # re-verify → expect all in-
                                   sync (exit 0)
```

Order note: run G12's generate_*_summary.sh FIRST (so the summary SOURCE is fresh), THEN docs_chain sync --all (so the html/pdf exports hash-match).

Effort: SMALL (2 commands, after G12). **Safe to auto-fix?** YES for the issues/fixed/governance contexts (mechanical export regen). The semgrep_status context belongs to the parallel semgrep session — coordinate before syncing it (§11.4.84 quiescence).

Quick-wins vs operator-gated

Quick-win (safe to auto-fix now, ~5 min total): - G12 →

generate_issues_summary.sh + generate_fixed_summary.sh (in-place). -

G14 → docs_chain sync --all --root . (run AFTER G12; coordinate semgrep_status with the parallel session).

Needs operator decision / judgment: - G8 → choose the closed-vocab Reason for HXC-044 (unsupported-topology proposed) OR petition to add not-reproducible to §11.4.90 vocab; then the one/two-line doc edit + Triple-check: token fix. - G7 → baseline-bump (exempt 116 pre-existing) vs author 117 transcripts. - G13 → 22 WebFetch-cross-referenced footers (subagent-parallelisable). - G1 → operator-scoped 64-submodule fleet cascade (propagate-governance.sh + backfill_anchor_cascade.sh + per-repo push), NOT run inline.

Captured-evidence index

- Live sweep summary: /tmp/sweep_full.out
- Per-gate: /tmp/g1-cascade.out /tmp/g7-qa.out /tmp/g8-obs.out /tmp/g12-summary.out /tmp/g13-sv.out /tmp/g14-verify.log
- G12 fix proven: in-place regen → --check PASS on both, diff = date-stamp only.
- G14 engine confirmed at ~/.local/bin/docs_chain + sibling ../docs_chain.

\$11.4.6: every count/date/sample above is verbatim from captured command output; nothing inferred. Read-only — no source/governance file edited; the transient G12 regen was reverted (git checkout -- docs/{Issues,Fixed}_Summary.md).